

Sarthak Kakkar

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Computer Science Student at Northeastern | AI/LLMs • Full-Stack Development • Embedded Systems

EDUCATION

Bachelor of Science in Computer Science | Business Administration minor

Northeastern University, Khoury College of Computer Science | Boston, MA | GPA: 3.72 | 08/2026

WORK EXPERIENCE

Fresenius Medical Care | Lawrence, USA | 01/2026 - 06/2026

Computer Science / Software Engineer Co-op | Full-time

- Built a high-accuracy **RAG knowledge system** on **Azure** over clinical and technical healthcare documents using specialized retrieval pipelines.
- Architected a **modular pipeline** to compare multiple RAG configurations, improving retrieval performance through evaluation metrics.
- Introduced agentic workflows like **LangGraph** for controlled, context-aware interactions and patient-facing AI systems.

Movement NeuroScience Lab | Boston, USA | 01/2025 - 06/2025

Embedded Systems Engineer Co-op | Full-time

- Engineered **C++/Python** embedded system with electrical stimulators & force sensors, enabling **IRB-approved** human subject experiments.
- Re-architected a vendor hardware API into an **asynchronous, modular system**, enabling real-time stimulation control across **3+ projects**.
- Automated end-to-end pipelines from raw sensor input to **clinician-ready visualizations**, producing **research-grade** reports for federal trials.

Wissen Infotech | Bangalore, India | 06/2024 - 08/2024

LLMOps Intern | Full-time

- Built **Python code-conversion accelerator** projected to reduce enterprise migration timelines by **2–3 months** across global client systems.
- Drove **LLM adoption** by delivering technical prototypes/demos to executives and prospective clients, influencing high-level business strategy.
- Designed **LangChain/LangGraph** feedback loop for autonomous error detection, targeted regeneration, and recovery, boosting pipeline efficiency.

Applify Tech Private Limited | Mohali, India | 07/2023 - 08/2023

Intern | Full-time

- Developed **Java backend services** with **REST APIs**, **SQL** queries, and **JDBC** integration, strengthening scalability and production reliability.
- Implemented **JaCoCo unit tests** and **debugged** production code, improving release stability and reducing runtime issues.
- Collaborated in an **agile** environment using **Git** and **JIRA**, documenting workflows and aligning sprint goals with client needs.

PROJECTS

PawtoGrader Platform

- Served as **Technical Lead and full-stack developer**, driving development with **TypeScript/Supabase/Docker** and owning project deliverables.
- Selected for an exclusive practicum project, **leading a 6-person Agile team** to extend PawtoGrader for **3k+ Khoury College CS students**.
- Delivered new features including **surveys, dashboards, and automated group formation**, improving platform usability and scalability.

AI Powered Communication Assistant

- Developed an **AI-powered communication assistant** using LangChain, LangGraph, and LangServe to automate professional inquiry handling.
- Deployed a **Dockerized FastAPI backend** on **AWS App Runner** for scalable, stateless processing.
- Integrated a **React + Vite frontend** hosted on **Netlify**, enabling secure, real-time interaction and notification delivery.

Retrieval as a Service

- Built and open-sourced **RaaS**, a self-hosted retrieval layer for RAG workflows with **project-scoped APIs** and **Chroma** indexing.
- Developed a **FastAPI + Next.js** system supporting document upload, ingestion, and secure per-project API key access.
- Designed a modular retrieval backbone with **Dockerized deployment** enabling reusable, scalable RAG pipelines without repeated setup.

SKILLS

Programming Languages: C++, Java, JavaScript, Python, SQL, TypeScript

Frameworks & Libraries: LangChain, LangGraph, Matplotlib, NumPy, Pandas, PyTorch, scikit-learn

Systems & Tools: AWS, Azure, Git, GitHub, JIRA, Linux, Netlify, Streamlit

Software Development: Agile, API Development, Async Programming, CI/CD, Docker, E2E/Unit Testing, Object-Oriented Design (OOD), PostgreSQL

Frontend & Web Development: HTML5, Next.js, Playwright Testing, React, React Router DOM, Server-Side Development, Tailwind CSS

Backend Development: Authentication & Authorization, Database Integration, Express.js, FastAPI, RESTful API Design, Server-Side Development